

Coding & Solution

Date	03-July-2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/s-ath-vika/Smart-Lender
Programming Language(s)	Python, HTML5, CSS3
Framework(s) Used	Flask (Python Web Server Gateway Interface)
Key Features Implemented	* Automated balancing of training instances using the Synthetic Minority Over-sampling Technique (SMOTE). * Real-time evaluation via a serialized Random Forest Classification model binary (rdf.pkl). * Data pipeline scaling consistency implemented via a serialized Scikit-Learn StandardScaler binary (scale1.pkl). * Modular, two-column responsive desktop input console equipped with responsive browser scaling media queries. * Custom contextual routing engine dynamically returning tailored responsive approval or disapproval layouts based on model results.

Pending / Incomplete Features	None. All functional application stages—from data scaling up to live web interface prediction rendering—are fully operational.
Setup / Run Instructions	1. Install Dependencies: Open terminal and run: pip install flask numpy==1.26.4 pandas scikit-learn imbalanced-learn 2. Train Model / Generate Pickles: Open VS Code, connect to the Anaconda base kernel, and select Run All inside loan.ipynb. This generates the rdf.pkl and scale1.pkl files directly into your workspace. 3. Run Web App: In your command prompt terminal, run: python app.py 4. Access Interface: Open your internet browser and navigate to http://127.0.0.1:5000 to access the prediction portal.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes